PROJECT REPORT

ON

SECURE HUMAN IDENTITY & LIVENESS EVALUATION DETECTION (SHIELD)

Carried Out at

CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

UNDER THE SUPERVISION OF

C-DAC

Submitted by

Sthavir Punwatkar

Mihir Rabade

PG CERTIFICATE PROGRAMME IN ARTIFICIAL INTELLIGENCE (PGCP-AI), CDAC GUWAHATI

C-DAC

Candidate's Declaration

We hereby certify that the work presented in this report entitled "Secure Human Identity & Liveness Evaluation Detection (SHIELD)", in partial fulfillment of the requirements for the award of the **PG Certificate Programme in Artificial Intelligence (PGCP-AI)**, and submitted to the **Centre for Development of Advanced Computing (C-DAC), Guwahati**, is an authentic record of our work carried out during the project period.

The matter presented in this report has not been submitted by us for the award of any degree of this or any other Institute/University.

Candidates:

Sthavir Punwatkar

Mihir Rabade

Acknowledgement

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Student Names:
Sthavir Punwatkar

Mihir Rabade

Abstract

With the rapid digital transformation of remote interviews, online examinations, and secure banking access, the threat of identity spoofing and presentation attacks has escalated significantly. Malicious actors increasingly employ high-resolution printed photos, replay video attacks, and sophisticated silicone masks to bypass traditional facial recognition systems. To combat these advanced threats, this project presents the **Secure Human Identity & Liveness Evaluation Detection (SHIELD)** system, a professional-grade, real-time multimodal liveness detection framework.

SHIELD utilizes a **hybrid approach** to differentiate genuine human presence from spoofing attempts by analyzing four independent biological and physical signals. The system combines **passive texture analysis** using a deep learning architecture (MiniFASNet) to identify non-human surface artifacts, **physiological verification** via a **3D Convolutional Neural Network (PhysNet)** for **Remote Photoplethysmography (rPPG)** to detect human pulse blood volume changes, and **behavioral consistency tracking** using real-time spatial facial landmarks. Furthermore, an **active challenge-response mechanism** validates temporal consistency, effectively mitigating deep-replay attacks.

The architecture comprises a **Flutter-based frontend client** communicating via **low-latency WebSockets** with a **synchronous FastAPI Python backend**. The backend orchestrates a complex inference pipeline utilizing **Ultralytics YOLOv8** for face detection, **MediaPipe** for mesh extraction, and **ONNX Runtime** for optimized deep learning model execution.

A rigorous testing and ablation study evaluated the system's performance, achieving an Attack Presentation Classification Error Rate (APCER) of 1.2% and an Average Classification Error Rate (ACER) of 1.0%, with an end-to-end inference latency of approximately 45ms per frame on local execution. The study also highlighted areas for future optimization, specifically decoupling synchronous database I/O from the ASGI event loop and streamlining the computer vision pipeline by deprecating redundant bounding box extractions. Overall, SHIELD establishes a robust foundation for building zero-trust, continuous authentication environments resilient to modern presentation attacks.

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Abbreviations & Acronyms

AI Artificial Intelligence

ML Machine Learning

DL Deep Learning

CNN Convolutional Neural Network

FAS Face Anti-Spoofing

rPPG Remote Photoplethysmography

APCER Attack Presentation Classification Error Rate

BPCER Bona Fide Presentation Classification Error Rate

ACER Average Classification Error Rate

API Application Programming Interface

JSON JavaScript Object Notation

ASGI Asynchronous Server Gateway Interface

ONNX Open Neural Network Exchange

SEB Safe Exam Browser

ROI Region of Interest

CV Computer Vision

CHAPTER 1: INTRODUCTION

1.1 Overview

The **Secure Human Identity & Liveness Evaluation Detection (SHIELD)** is a state-of-the-art liveness detection system designed to authenticate genuine human presence in digital environments. As remote interactions become the standard, systems must robustly differentiate between a real user and a presentation attack (PA), such as a printed photograph or a digital screen replay. SHIELD addresses this by orchestrating a highly optimized inference pipeline that evaluates texture, pulse (rPPG), and behavioral biometrics simultaneously, providing an explainable and decisive verdict through a unified fusion engine.

1.2 Background

Traditional facial recognition systems focus solely on matching geometric facial features to a stored template, answering the question "Who is this?" rather than "Is this a live person?". This limitation has been aggressively exploited using Presentation Attacks (PAs). A Presentation Attack Instrument (PAI) can range from simple 2D paper masks and smartphone screens displaying videos to complex 3D silicone masks. Developing countermeasures, known as Presentation Attack Detection (PAD) or Face Anti-Spoofing (FAS), has become an active field of research spanning computer vision and signal processing.

1.3 Motivation

The motivation behind SHIELD stems from the vulnerabilities observed in current virtual environments such as Remote Interview Verification, Automated Attendance Systems, and Remote Proctoring. High-stakes assessments face constant threats from impersonation. Many existing solutions rely on unimodal passive texture analysis, which can be bypassed by high-resolution digital screens. A system is needed that not only observes the physical texture of the face but confirms underlying biological processes, such as the micro-color changes in the skin corresponding to the human heartbeat.

1.4 Problem Statement

Existing unimodal liveness detection systems suffer from high False Acceptance Rates (FAR) when subjected to sophisticated replay or 3D mask attacks. Furthermore, systems that incorporate multiple checks often suffer from prohibitive latency, rendering them unusable for real-time video streaming authentication. The challenge is to design and implement a multimodal liveness detection framework that seamlessly fuses deep learning texture analysis, physiological blood flow verification (rPPG), and temporal behavioral tracking into a single, low-latency pipeline (under 100ms) without degrading the user experience.

CHAPTER 2: LITERATURE SURVEY

Automated Face Anti-Spoofing (FAS) has transitioned from traditional hand-crafted feature extraction to advanced spatial-temporal deep learning analysis. Early implementations relied heavily on texture descriptors such as Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG) [1]. While computationally efficient, these models struggled to generalize across diverse camera sensors and illumination environments.

The advent of Convolutional Neural Networks (CNNs) revolutionized FAS. Single-shot object detection architectures such as YOLO [2] enabled rapid face localization, while lightweight CNNs based on MobileNet paradigms successfully traded marginal accuracy drops for massive latency improvements on edge devices [3]. The MiniFASNet architecture, employed by SHIELD, builds upon these principles to detect high-frequency spoofing artifacts like moiré patterns with sub-millisecond latency.

Simultaneously, Remote Photoplethysmography (rPPG) emerged as a critical biological liveness indicator, positing that synthetic masks and photographs fundamentally lack a biological pulse [4]. Recent research introduced 3D Convolutional Neural Networks (PhysNet) capable of extracting spatiotemporal pulse signals directly from raw video frames [5].

Furthermore, behavioral biometrics leveraging facial meshes, such as Google's MediaPipe [6], provide geometric verification of liveness through blink detection and head pose estimation.

Despite these advancements, unimodal systems remain vulnerable to specific Presentation Attack Instruments (PAIs). Consequently, fusion strategies integrating texture, physiological, and behavioral signals are now required to meet ISO/IEC 30107-3 standards [7]. The primary research gap exists in minimizing the computational overhead of fusing 3D-CNNs and 2D-CNNs in real-time server environments. SHIELD resolves this via an asynchronous, threshold-gated cascade architecture.

Table 2.1: Research Paper Comparison

Topic / Paper	**Methodology**	**Advantages**	**Limitations**	**Relation to SHIELD**
YOLOv8 [8]	Anchor-free single-stage detection.	High bounding box accuracy and speed.	Computationally heavy for simple crops.	Used for initial spatial ROI extraction.
MediaPipe Mesh [6]	ML-based 468-point 3D facial topology.	Sub-5ms latency, cross-platform.	Susceptible to extreme yaw angles.	Drives Behavioral Analyzer (EAR/MAR).
MiniFASNet	Lightweight CNN for texture anomaly detection.	Optimized for mobile/edge inference.	Vulnerable to high-fidelity 3D masks.	Core Anti-Spoofing Engine for 2D attacks.
PhysNet rPPG [5]	3D-CNN processing facial video volumes.	Highly robust to physical masking.	Requires strict lighting and temporal buffering.	Physiological rPPG Detector for pulse validation.

CHAPTER 3: OBJECTIVES

3.1 Project Objective

To develop and deploy a professional-grade, multi-modal Presentation Attack Detection (PAD) framework capable of authenticating genuine human presence in zero-trust remote environments using real-time video streams.

3.2 Specific Objectives

- **Technical Objectives:** To integrate YOLOv8, MediaPipe, and MiniFASNet into a synchronized, threshold-gated inference pipeline achieving sub-100ms end-to-end latency per frame.
- **Security Objectives:** To detect and neutralize 2D printouts, digital video replays, and 3D silicone masks by enforcing passive texture analysis and active behavioral challenge-response mechanisms.
- **Implementation Objectives:** To establish a low-latency bidirectional WebSocket architecture bridging a Flutter client and a FastAPI python backend, backed by SQLite for session state persistence.
- **Research Objectives:** To quantify the latency impact of redundant cascaded object detectors (e.g., YOLOv8 followed by MediaPipe) and evaluate CPU-bound ONNX execution for 1D-CNN rPPG extraction.

CHAPTER 4: SCOPE

4.1 Current Scope

The SHIELD system encompasses a Flutter-based mobile/desktop client for video capture and a FastAPI server for backend inference. The implemented pipeline supports real-time Face Detection, 468-point landmark extraction, passive texture analysis (MiniFASNet), and active behavioral challenges. Session telemetry is securely persisted to a local SQLite database.

4.2 Applications & Target Users

SHIELD is targeted toward high-stakes remote environments including Remote Proctoring, automated Know Your Customer (KYC) banking pipelines, and secure corporate access control. Target users include educational institutions and financial entities requiring stringent identity verification without physical human oversight.

4.3 Advantages vs Limitations

Table 4.1: Advantages vs Limitations

Implemented Advantages	**Current Limitations**
Highly optimized ONNX Runtime execution for CPU infrastructure.	CPU-bound ML inference executes synchronously on the ASGI event loop.
Cascaded thresholding aborts inference early on obvious spoofs, saving compute.	SQLite database-level locking causes bottlenecking under high concurrent client loads.
Multimodal fusion prevents unimodal bypass attacks.	rPPG extraction requires strict ambient lighting constraints.

4.4 Future Scalability & Deployment Scope

Future iterations will migrate the monolithic inference logic to distributed Celery/Redis worker queues and transition the datastore from SQLite to PostgreSQL. This decoupled asynchronous architecture will unlock massive horizontal scalability. Furthermore, the repository indicates a planned deprecation of YOLOv8 in favor of utilizing existing MediaPipe data for ROI extraction to further streamline the computer vision pipeline.

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CHAPTER 5: SYSTEM ARCHITECTURE

SHIELD operates on a client-server paradigm, heavily utilizing WebSockets for real-time telemetry and video streaming. The architecture prioritizes separation of concerns, routing raw frames from the Flutter client to the FastAPI backend, where they enter the synchronous Computer Vision (CV) pipeline.

5.1 Overall System Architecture

The user interacts with the Flutter application, which accesses the device camera. Frames are encoded and dispatched via WebSockets. The FastAPI backend decodes the payload and routes the raw Numpy matrices to the Fusion Service, which orchestrates the AI models and records the verdict in the SQLite database.

5.2 Inference Pipeline Data Flow

A single frame is processed by YOLOv8 for spatial bounding boxes and MediaPipe for dense facial landmarks. The cropped Region of Interest (ROI) is passed to the Mini-FASNet model to detect texture irregularities. Concurrently, the rPPG network analyzes the crop for pulse variations, and the Behavioral analyzer computes Eye Aspect Ratios (EAR). The Fusion Engine aggregates these multi-dimensional vectors to determine if the frame is a Live Human or a Presentation Attack.

5.3 Request Lifecycle & Concurrency

Every frame enters the pipeline and is immediately gated by a Quality Check. Blurry or poorly lit frames are rejected to prevent poisoning the temporal buffers of the rPPG models. Valid frames undergo spatial analysis (texture extraction) and temporal analysis (pulse and behavior tracking). The final aggregated result is committed to the database and streamed back to the client UI. As highlighted in the Root Cause Investigation, the backend currently executes the entire CV pipeline synchronously within the WebSocket event loop, presenting a bottleneck requiring migration to asynchronous workers.

CHAPTER 6: TECHNOLOGIES USED

6.1 Frontend Technologies

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Figure 6.1: Frontend Architecture

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Figure 6.1: Frontend Architecture

Flutter: Flutter was selected as the cross-platform UI toolkit. Its isolate-based concurrency model allows the heavy computational task of camera stream encoding to run independently from the main UI thread, ensuring smooth 60 FPS animations. Alternatives like React Native were considered but discarded due to the JavaScript bridge overhead during high-frequency frame capture.

6.2 Backend & Server Technologies

FastAPI & Python: FastAPI serves as the backend framework, selected for its native ASGI asynchronous support and seamless WebSocket integration, which are critical for real-time streaming architectures. Python acts as the glue language given its unparalleled ecosystem for computer vision and machine learning.
WebSockets: Utilized for continuous bi-directional telemetry. Standard REST HTTP

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Figure 6.2: Backend Architecture

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Figure 6.2: Backend Architecture

was avoided due to the unacceptable latency overhead of establishing new TCP handshakes for every video frame.

6.3 Computer Vision & Machine Learning

OpenCV & NumPy: OpenCV handles raw image matrix transformations, color space conversions, and geometric cropping. NumPy provides the underlying C-optimized tensor data structures required by the ML models.
MediaPipe: Selected to provide dense 468-point 3D facial meshes at sub-5ms latency. This drives the Behavioral Analyzer without relying on heavy neural networks.
Ultralytics YOLOv8: Utilized for highly accurate spatial bounding box extraction.
ONNX Runtime: The PyTorch MiniFASNet and rPPG models were exported to the Open Neural Network Exchange (ONNX) format. ONNX Runtime was explicitly selected to allow the backend to utilize the CPUExecutionProvider for high-speed inference, removing strict hardware dependencies on NVIDIA CUDA GPUs.

6.4 Database & Deployment

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Figure 6.3: Deployment Diagram

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Figure 6.3: Deployment Diagram

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Figure 6.4: Docker Architecture

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Figure 6.4: Docker Architecture

SQLite & SQLAlchemy: SQLite is currently implemented for rapid prototyping and local session persistence via SQLAlchemy ORM. However, its file-level locking is a known limitation.
Docker: Docker and Docker Compose containerize the application, ensuring environmental parity between development and production deployments.

Table 6.1: Technology Selection Matrix

Domain	**Selected Tech**	**Alternatives Considered**	**Primary Advantage**
Frontend UI	Flutter	React Native	Isolate-based threading for camera streams.
Backend API	FastAPI	Django, Flask	Native WebSocket and ASGI support.
Face Mesh	MediaPipe	dlib, MTCNN	Sub-5ms latency on CPU.
Model Execution	ONNX Runtime	PyTorch Native	Hardware-agnostic CPU optimization.

CHAPTER 7: FUNCTIONAL MODULES

The SHIELD repository is strictly categorized into specific functional modules across the client and server.

7.1 Frontend Modules (Flutter)

- **CameraCaptureService:**
  Location: `frontend/lib/services/camera_capture_service.dart`
  Purpose: Interfaces with platform camera APIs to capture raw frames.
  Responsibilities: Grabs frames at 30 FPS and pushes them to the transport queue.

- **FrameTransportService:**
  Location: `frontend/lib/services/frame_transport_service.dart`
  Purpose: Manages network backpressure and payload transmission.
  Responsibilities: Compresses frames into H.264 byte chunks and transmits them via WebSockets interlaced with JSON metadata.

- **ChallengeService:**
  Location: `frontend/lib/services/challenge_service.dart`
  Purpose: Manages the active behavioral challenge state machine.
  Responsibilities: Parses backend commands, runs local UI countdown timers, and emits state changes to the `challenge_screen.dart` UI.

7.2 Backend Orchestration Modules

- **StreamingDecoder:**
  Location: `backend/services/video_decoder.py`
  Purpose: Unpacks network payloads.
  Responsibilities: Decodes H.264 binary WebSocket chunks back into raw `numpy` BGR matrices for the ML pipeline.

- **SessionManager:**
  Location: `inference/session_manager.py`
  Purpose: Manages active verification sessions.
  Responsibilities: Aggregates challenge states, tracks historical predictions for temporal consistency, and handles timeouts.

- **LocalDBService:**
  Location: `backend/services/db_service.py`
  Purpose: Data persistence.
  Responsibilities: Stores session metadata and final verdicts into the `shield_local.db` SQLite database.

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Figure 7.2: API Communication

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Figure 7.1: API Communication

7.3 AI & Inference Modules

- **QualityScoreEngine:**
  Location: `inference/quality/`
  Purpose: Evaluates frame viability.
  Responsibilities: Calculates brightness, blur, and contrast. Rejects frames to prevent noisy data from polluting the temporal rPPG buffers.

- **YoloSegDetector:**
  Location: `inference/yolo_detector.py`
  Purpose: Spatial localization.
  Responsibilities: Executes YOLOv8 to extract facial bounding boxes and masks. *Future Improvement:* Deprecate in favor of MediaPipe bounding boxes to eliminate a 28ms latency penalty.

- **BehavioralAnalyzer:**
  Location: `inference/behavioral_analyzer.py`
  Purpose: Tracks dynamic facial geometry.
  Responsibilities: Extracts 468 landmarks via MediaPipe. Computes the Eye Aspect Ratio (EAR) for blink detection and Perspective-n-Point (PnP) for head pose.

- **AntispoofInference:**
  Location: `inference/antispoof/inference.py`
  Purpose: Defends against 2D printed and digital spoofs.
  Responsibilities: Normalizes the cropped ROI and executes the `minifas_antispoof_v2.onnx` model. Outputs a spatial spoof probability.

- **RPPGDetector:**
  Location: `inference/rppg_detector.py`
  Purpose: Physiological pulse verification.
  Responsibilities: Extracts the green channel signal across a temporal buffer of spatial crops. Executes a 1D-CNN over the time-series data to estimate blood flow volumetric changes.

- **FusionEngine:**
  Location: `inference/fusion_engine.py`
  Purpose: Final decision orchestration.
  Responsibilities: Applies weighted cascade logic to the incoming multi-modal scores. Defaults to 'Spoof' if the aggregated threshold falls below 0.5.

Table 7.1: Repository Module Summary

Module	**Type**	**Primary Output**
FrameTransportService	Frontend	H.264 Byte Stream
StreamingDecoder	Backend	Numpy BGR Matrices
YoloSegDetector	AI Inference	Bounding Box Coordinates
BehavioralAnalyzer	AI Inference	EAR, MAR, Head Pose Vectors
AntispoofInference	AI Inference	2D Texture Liveness Score
RPPGDetector	AI Inference	3D Physiological Pulse Score
FusionEngine	Backend	Final Boolean Verdict

CHAPTER 8: SYSTEM REQUIREMENTS

8.1 Hardware Requirements

- **Server:** Minimum 4 CPU Cores (8 recommended for concurrent ONNX inference), 8 GB RAM. (GPU not strictly required due to ONNX CPU optimization, but NVIDIA CUDA support significantly increases throughput).
- **Client (Mobile/Desktop):** Modern smartphone or Web Browser with a 720p minimum webcam.

8.2 Software Requirements

- **Backend:** Python 3.10+, FastAPI, ONNXRuntime, OpenCV-Python, Ultralytics, MediaPipe.
- **Frontend:** Flutter SDK 3.19+, Dart.
- **Database:** SQLite3 (Local) / PostgreSQL 14+ (Production).
- **Deployment:** Docker, Docker Compose (for scalable deployments).

CHAPTER 9: AI PIPELINE & ALGORITHMS

9.1 Overall AI Pipeline

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Figure 8.1: Overall System Architecture

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Figure 9.1: Overall System Architecture

The SHIELD inference pipeline operates on an asynchronous, threshold-gated cascade. When a client application streams an H.264 video payload, the backend decodes it into a sequence of NumPy BGR matrices. These frames enter the pipeline and are immediately subject to Quality Validation. If ambient lighting is insufficient, the frame is discarded to preserve temporal integrity. Valid frames proceed to spatial extraction (YOLOv8 + MediaPipe), passive texture analysis (MiniFASNet), and physiological pulse extraction (rPPG). The individual model confidences are subsequently aggregated by the Fusion Engine. For a visual representation, refer to the generated *Figure 8.1: Overall System Architecture* diagram in the architecture documentation.

9.2 Face Detection

Purpose: To spatially localize the user's face and extract the primary Region of Interest (ROI) for downstream models.

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Figure 8.2: Data Flow Diagram (Level 1)

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Figure 9.2: Data Flow Diagram (Level 1)

Algorithm & Implementation: The repository implements Ultralytics YOLOv8 (`yolov8n-seg.pt`). The model accepts a downscaled 640×640 tensor and outputs a bounding box tensor (x, y, w, h) along with object class confidence.
Advantages: Single-shot detection provides unparalleled accuracy against complex backgrounds.
Limitations: Profiling the repository revealed that executing YOLOv8 alongside MediaPipe introduces a redundant 28ms latency penalty, making it a target for future architectural deprecation.

9.3 Face Mesh & Alignment

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Figure 8.3: Inference Pipeline

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Figure 9.3: Inference Pipeline

Algorithm & Implementation: The repository utilizes the Google MediaPipe FaceLandmarker task. The model projects a dense 468-point 3D facial topology onto the 2D bounding box. These geometric points drive the behavioral tracker and allow for tight, precise cropping of the forehead and cheeks for the rPPG pipeline. Refer to *Figure 8.2: Inference Pipeline* for the sequence flow.

9.4 Blink Detection

Purpose & Implementation: To satisfy active challenge-response protocols (e.g., "Blink"), the Behavioral Analyzer calculates the Eye Aspect Ratio (EAR) dynamically across consecutive frames. **Mathematical Logic:** The EAR equation leverages the Euclidean distance between the vertical (p_2, p_6 and p_3, p_5) and horizontal (p_1, p_4) eye landmarks:

$$EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2||p_1 - p_4||} \tag{9.1}$$

If $EAR < \tau_{blink}$ (where the threshold $\tau_{blink} \approx 0.2$) for a specified temporal window, a blink event is registered and verified against the active state machine.

9.5 Head Pose Estimation

Algorithm: To detect active "Look Left" or "Look Right" challenges, the system calculates the geometric head pose using the Perspective-n-Point (PnP) algorithm. **Implementation:** By mapping the 2D MediaPipe landmarks to a generic 3D human face model, the system solves the extrinsic camera matrix to extract rotation vectors. These are converted into Euler Angles:

- **Yaw (θ):** Left/Right rotation.
- **Pitch (ϕ):** Up/Down rotation.
- **Roll (ψ):** Tilt.

The `BehavioralAnalyzer` asserts challenge success if the Yaw exceeds a threshold of $\pm 25°$.

9.6 MiniFASNet Anti-Spoofing

Purpose: To perform passive texture analysis and detect surface-level presentation attacks (e.g., digital screens, paper masks).
Implementation: The system executes `minifas_antispoof_v2.onnx` via ONNXRuntime.
Inference Pipeline: The cropped facial ROI is normalized into a 80×80 tensor. MiniFASNet utilizes depthwise separable convolutions to extract high-frequency features such as moiré patterns. A final Softmax layer converts the logits into a normalized probability:

$$P(y = live \mid x) = \frac{e^{z_{live}}}{e^{z_{live}} + e^{z_{spoof}}} \tag{9.2}$$

Strengths: Exceptionally fast (sub-10ms on CPU).
Limitations: Susceptible to highly realistic 3D silicone masks which do not present 2D surface artifacts.

9.7 Remote Photoplethysmography (rPPG) Pipeline

Purpose: To detect volumetric changes in facial blood flow representing a live human pulse.
Implementation: The `RPPGDetector` module enforces a strict signal processing pipeline.

1. **ROI Selection:** The cheeks and forehead are isolated utilizing MediaPipe coordinates.
2. **Signal Extraction:** The spatial mean of the Green channel is extracted per frame, as hemoglobin absorption peaks in the green spectrum.
3. **Temporal Buffering:** A sliding window of N frames is maintained.
4. **Filtering:** A digital band-pass filter ($[0.7\text{Hz}, 3.0\text{Hz}]$) removes low-frequency lighting changes and high-frequency noise.
5. **Inference:** The filtered 1D time-series signal is passed through a 1D-CNN (`rppg_1dcnn_v2.onnx`), outputting a pulse confidence score.

9.8 Feature Fusion Engine

Logic: The `FusionEngine` aggregates the independent modal scores into a single verdict.
Implementation: SHIELD utilizes a weighted sum fusion approach. During a passive monitoring state (no active challenge), the system applies predefined weights:

$$Score_{final} = w_{app} \cdot S_{MiniFASNet} + w_{rPPG} \cdot S_{pulse} + w_{beh} \cdot S_{dynamic} \tag{9.3}$$

Where $w_{app} = 0.5$, $w_{rPPG} = 0.3$, and $w_{beh} = 0.2$. If $Score_{final} < 0.5$, the frame is classified as a Spoof.

9.9 Decision Engine

The `SessionManager` enforces the final authentication rules. If the FusionEngine classifies an attack, or if a user fails to complete a randomized challenge within the temporal timeout window (typically 10 seconds), the session state transitions to `Rejected`. Database logging occurs simultaneously.

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`Figure_8_4_Fusion_Engine.pdf`

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Caption:
Figure 8.4: Fusion Engine

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Figure 9.4: Fusion Engine

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`Figure_8_5_Activity_Diagram.pdf`

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Figure 8.5: Activity Diagram

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Figure 9.5: Activity Diagram

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`Figure_8_6_Session_State_Diagram.pdf`

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Figure 8.6: Session State Diagram

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Figure 9.6: Session State Diagram

Table 9.1: AI Model Summary

Model	**Type**	**Inputs**	**Outputs**
YOLOv8-Seg	2D CNN	$640 \times 640 \times 3$ Tensor	Bounding Box & Mask
MediaPipe Mesh	MLP Cascade	Bounding Box ROI	468 3D Landmarks
MiniFASNet	Lightweight CNN	$80 \times 80 \times 3$ Tensor	Softmax Liveness Score
1D-CNN rPPG	Temporal CNN	1D Pulse Array (Time-series)	Biological Liveness Score

CHAPTER 10: SECURITY & LIVENESS DETECTION

10.1 Threat Model

Presentation Attacks (PAs) aim to subvert biometric authentication by presenting an artifact to the camera sensor. The SHIELD threat model encompasses:

- **Printed Photo Attacks:** A high-resolution 2D image presented to the lens.
- **Screen Replay Attacks:** A pre-recorded video or photograph displayed on a digital screen (e.g., iPad, smartphone).
- **3D Mask Attacks:** Silicone, paper, or resin masks worn over a human face.
- **Deepfake/Virtual Camera Injections:** Synthetically generated face swaps injected directly into the video feed at the OS level.

10.2 Defense Mechanisms

10.2.1 Passive Texture Analysis (MiniFASNet)

Digital screens emit distinct chromatic profiles and moiré patterns due to pixel grids, while printed photos lack specular reflections. MiniFASNet excels at identifying these high-frequency artifacts.
Limitation: A high-quality 3D silicone mask mimics human skin texture, often bypassing 2D convolutional filters.

10.2.2 Physiological Liveness (rPPG)

To counter 3D masks, SHIELD employs rPPG. Synthetic artifacts, regardless of texture quality, do not possess a cardiovascular system. By strictly filtering and classifying the micro-variations in the green color channel of the facial ROI, SHIELD accurately identifies the lack of a human pulse.
Limitation: The rPPG signal is extremely susceptible to ambient lighting noise and severe head motion.

10.2.3 Dynamic Behavioral Tracking

To defeat static photos and pre-recorded videos, the `ChallengeService` prompts the user with randomized tasks ("Look Left", "Smile"). The `BehavioralAnalyzer` tracks the Euler angles and MediaPipe landmarks to verify intent in real-time.

10.2.4 Environment Security

To defend against virtual camera injections (e.g., OBS Virtual Cam), the system enforces Safe Exam Browser (SEB) cryptographic header validation. The FastAPI backend rejects all WebSocket handshakes lacking a valid SEB trust token, ensuring the video feed originates from a secure hardware boundary.

Table 10.1: Threat vs Defense Matrix

Attack Vector	**Primary Defense Mechanism**	**Effectiveness**
Printed Photograph	MiniFASNet, Behavioral Challenge	Very High
Screen Replay	MiniFASNet, rPPG (No pulse)	Very High
3D Silicone Mask	rPPG (No pulse), Behavioral Challenge	High
Virtual Camera Swap	SEB Token Validation, HTTP Headers	Absolute (By Design)

Table 10.2: Liveness Technique Comparison

Technique	**Strengths**	**Computational Overhead**
Texture Analysis (2D)	Instantaneous decision, highly robust to screens.	Low (CPU Optimized via ONNX)
Physiological (rPPG)	Unmatched against 3D masks and high-fidelity prints.	High (Requires sliding window buffering)
Behavioral Response	Highly interactive, defeats static replays.	Medium (Requires dense facial landmarking)

10.3 Conclusion on System Security

SHIELD implements a defense-in-depth architecture. By aggregating passive spatial metrics, active temporal behavior, and physiological pulse detection through a unified Fusion Engine, the system mathematically ensures that bypassing one modality (e.g., using a high-res screen to bypass rPPG) triggers a critical failure in another (MiniFASNet detecting screen borders).

CHAPTER 11: TESTING, VALIDATION & PERFORMANCE EVALUATION

11.1 Testing Strategy

The SHIELD system employs a multi-tiered testing methodology designed to rigorously evaluate the real-time inference capabilities and security boundaries of the architecture.

- **Unit Testing:** Validates individual utility functions and model tensor shapes independently.
- **Integration Testing:** Ensures the seamless bridging of the Flutter WebSocket stream with the FastAPI inference pipeline.
- **End-to-End Testing:** Validates the complete request lifecycle from camera capture to final SQLite database persistence.
- **Performance Profiling:** Utilizes `cProfile` to identify latency bottlenecks across the AI cascade.
- **Component Validation:** Verifies the isolated behavior of sub-modules like the `ChallengeService` and `RPPGDetector`.

These strategies were selected to explicitly isolate AI model accuracy from network engineering stability, ensuring that both domains function flawlessly in real-time constraints.

11.2 Module Validation

Every major repository module underwent specific validation to verify structural integrity.

- **Flutter CameraCaptureService:** Verified for 30 FPS raw capture without dropping UI frames.
- **Flutter FrameTransportService:** Evaluated under severe network throttling. Expected backpressure handling was observed via frame dropping to maintain synchronization.
- **FastAPI StreamingDecoder:** Successfully unpacks H.264 byte chunks into valid Numpy BGR matrices. Corrupt binary payloads are caught gracefully.
- **QualityScoreEngine:** Successfully rejects completely black or entirely blurred images before passing them to the AI pipeline.
- **YOLOv8 & MediaPipe:** Bounding box extractions align perfectly with the 468-point facial mesh.
- **MiniFASNet Engine:** Spoof probabilities output correctly normalized floats between 0.0 and 1.0.

- **RPPGDetector:** Generates localized pulse spikes in well-lit environments. Validation failed under low-light conditions, exposing a known limitation.
- **FusionEngine:** Weights are applied correctly based on the active challenge state machine.

11.3 API & Communication Testing

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Figure 10.1: WebSocket Sequence Diagram

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Figure 11.1: WebSocket Sequence Diagram

The core communication protocol relies heavily on WebSockets due to the latency requirements of streaming video.

- **WebSocket Connection:** Verified persistent bi-directional telemetry. Connections drop cleanly upon client disconnect.
- **Authentication Validation:** API security successfully rejects unauthorized client requests that lack the correct Safe Exam Browser (SEB) cryptographic trust headers.
- **Timeout Behavior:** If no frames are received for 10 consecutive seconds, the server forcefully transitions the session to a `Rejected` state and closes the socket.
- **REST Endpoints:** *Not evaluated in the current implementation* as the primary lifecycle relies exclusively on WS pipelines.

11.4 AI Model Validation

The core AI components were evaluated against a combined corpus of the CASIA-FASD and CelebA-Spoof validation datasets. The quantitative metrics mathematically define the system's accuracy:

- **APCER (Attack Presentation Classification Error Rate):** Achieved 1.2%. This is well below the industry standard constraint of $< 5.0\%$, indicating a high resistance to successful spoof attempts.
- **BPCER (Bona Fide Presentation Classification Error Rate):** Achieved 0.8%. This easily beats the industry standard constraint of $< 3.0\%$, ensuring genuine users are not frequently rejected.
- **ACER (Average Classification Error Rate):** Conclusively established at 1.0%.

While MiniFASNet demonstrated exceptional accuracy against 2D prints and screen replays, the validation highlighted that high-fidelity 3D masks required the rPPG signal to successfully trigger a rejection.

11.5 System Performance

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`Figure_10_2_Execution_Timeline.pdf`

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Figure 10.2: Execution Timeline

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Figure 11.2: Execution Timeline

Performance was rigorously profiled utilizing `cProfile` and custom ablation scripts on test videos.

- **Inference Latency:** The end-to-end full pipeline latency clocks at ~ 45.45ms per frame, resulting in an effective processing throughput of ~ 22 FPS.
- **Ablation Study (YOLOv8 Penalty):** Profiling revealed that executing YOLOv8 for bounding boxes immediately prior to MediaPipe introduces a redundant 28ms execution penalty. Future iterations propose deprecating YOLOv8 to improve latency by approximately 40%.
- **Concurrency Bottleneck:** Simulated multiple concurrent WebSocket clients streaming 30 FPS video to the FastAPI server. The system exhibited severe bottlenecking under multi-client loads due to synchronous SQLite `cursor.execute` operations blocking the asynchronous ASGI thread.

GPU scaling metrics were *Not evaluated in the current implementation*, as the ONNXRuntime was explicitly configured for CPU execution providers.

11.6 Security Validation

- **Photo & Screen Replay Attacks:** Implemented and successfully blocked by the MiniFASNet spatial texture analysis.
- **Video Replay Attacks:** Implemented and blocked via randomized active behavioral challenges.
- **Virtual Camera Swap (Deepfakes):** Implemented and blocked at the network level via SEB header validation.
- **Face Mismatch:** *Future Work.* Advanced identity embeddings (e.g., MobileFaceNet) are proposed for future iterations.

11.7 Failure Analysis & Edge Cases

- **Environmental Lighting:** rPPG extraction fails fundamentally in poor ambient lighting, as the microscopic skin color variations are obscured by camera ISO noise.
- **Geometric Occlusion:** Identity and behavioral tracking leverages 2D Euclidean distances. Extreme head yaw rotations cause landmark occlusion, resulting in false identity rejections.
- **Concurrency:** As previously noted, the synchronous SQLite database locking mechanism prevents vertical scaling.
- **Multiple Faces:** *Not evaluated in the current implementation.* The pipeline assumes a single primary actor.

11.8 Results & Observations

SHIELD successfully fulfills its primary objective: executing a multimodal fusion cascade at sub-50ms latency with an ACER of 1.0%. The user experience remains seamless due to the decoupling of the Flutter UI thread and the camera isolation. While the system architecture is robust against physical presentation attacks, the primary limitations lie purely in software engineering concurrency (SQLite thread locking) and environmental constraints (low-light rPPG failure).

Table 11.1: Module vs Testing Method

Module	**Testing Method**	**Outcome**
Flutter WebSockets	Integration Testing	Passed (Robust reconnection)
FastAPI Decoder	Unit Testing	Passed (Decodes H.264)
MiniFASNet	Benchmark Validation	Passed (1.0% ACER)
SessionManager	E2E Validation	Passed (State transitions sync)

Table 11.2: Failure Scenario vs System Response

Failure Scenario	**System Response**
Network Disconnect	Session transitions to timeout; SQLite logs failure.
Extreme Head Yaw	MediaPipe loses landmarks; Quality Engine drops frame.
Low Ambient Light	Quality Engine drops frame; Prevents rPPG corruption.
Multiple Concurrent Clients	ASGI event loop stalls due to SQLite thread locking.

CHAPTER 12: FUTURE ENHANCEMENTS

- **Decoupled Asynchronous Architecture:** Migrating from a synchronous FastAPI + SQLite model to a distributed architecture using PostgreSQL, Redis, and Celery workers.
- **Computer Vision Streamlining:** Deprecating the YOLOv8 model in favor of utilizing the existing MediaPipe FaceLandmarker data for bounding box and ROI extraction.
- **Robust Face Alignment:** Implementing a 5-point Affine Transformation utilizing the eye and mouth coordinates prior to passing the ROI to MiniFASNet.
- **Advanced Identity Embeddings:** Replacing the current 2D-Euclidean geometric identity tracking with a lightweight facial embedding model (e.g., MobileFaceNet).
- **Edge Deployment (Mobile NPU):** Transitioning the ONNX models directly into the Flutter application utilizing TFLite or platform-specific ML APIs.

CHAPTER 13: CONCLUSION

The **Secure Human Identity & Liveness Evaluation Detection (SHIELD)** system successfully demonstrates the feasibility of real-time, multimodal presentation attack detection. By synthesizing deep-learning texture analysis (MiniFASNet), physiological pulse verification (rPPG), and dynamic behavioral tracking, the system effectively neutralizes a wide spectrum of spoofing vectors, from 2D printouts to digital replay attacks.

The architecture achieves impressive benchmark metrics (ACER of 1.0%) while maintaining a sub-50ms inference profile per frame. More importantly, rigorous forensic profiling and ablation studies conducted during development provided critical insights into computational bottlenecks. Identifying the redundancy of cascaded object detectors (YOLO + MediaPipe) and the concurrency limitations of synchronous database I/O paves the way for highly scalable future iterations.

Ultimately, SHIELD establishes a formidable foundation for next-generation biometric authentication, ensuring that remote digital environments remain secure, trustworthy, and resilient against evolving artificial intelligence and spoofing threats.